

Dhruv Nirmal

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PROFESSIONAL SUMMARY

Data Analyst & Engineer with 3+ years turning operational data from multiple systems into performance metrics, KPI benchmarks, and AI-augmented workflows - directly applicable to Melbourne Water's Works Delivery analytics mandate. My core domain is work-order-level transaction data, supplier and service-provider performance tracking, cost-leakage detection, and exception reporting: the same analytical discipline that underpins maintenance performance and reliability analytics. Experienced delivering production-grade Power BI, Tableau and Snowflake / SQL Server solutions; have shipped two agentic AI workflows (multi-agent data QA + automated report generation) and built predictive models on industrial/operational data - a rare combination in a BI candidate.

KEY SKILLS

Analytics & BI: Power BI, Tableau, Snowflake, SQL Server, Data Storytelling, KPI Analytics, Dashboards, Benchmarking, Exception Reporting

Data Engineering & QA: Python, SQL, ETL Pipelines, Data Cleaning, Data Validation, Anomaly Detection, Data Quality Controls, KNIME

AI & Automation: LLM Workflows (Claude, ChatGPT), Agentic AI, Workflow Automation (Python, MS Graph API), Automated Reporting, Process Streamlining

Cloud & Infrastructure: Azure, AWS (S3, EC2, IAM), Databricks, Terraform, Docker

Statistics & Modelling: Statistical Methods, Predictive Modelling, Forecasting, Logistic Regression, Scikit-learn

Soft Skills: Stakeholder Collaboration, Client-Facing Delivery, Cross-functional Communication, Problem Framing

EXPERIENCE

Data Analyst & Engineer, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 - Present | Melbourne, Australia

- Owned end-to-end analytics pipeline for the company's largest client - ~6M work-order-level transaction rows plus PO and lookup datasets; standardised data-capture formats across 13 external data submitters (parallel to improving data quality from maintenance service providers), replaced a manual KNIME process with fully automated SQL Server ingestion (Microsoft Graph API), saving ~3–3.5 hours weekly.
- Implemented a data QA program: automated row/column counts, data-type checks, and discrepancy logging that flag integrity issues and email client and internal teams for review - structurally identical to monitoring data-quality from external service providers in a Works Delivery context.
- Delivered 16+ Tableau dashboards enabling stakeholder teams to monitor supplier and service-provider performance, track KPIs, identify cost-leakage trends, and act on exception data across multiple datasets.
- Shipped a multi-agent AI workflow (5 agents + orchestrator, Python, Claude) to validate client datasets prior to delivery - a production QA control, not a prototype, catching data integrity issues systematically before they reached stakeholders.
- Deployed agentic AI workflows (Claude, ChatGPT) to auto-generate structured client performance reports and streamline analytics-script development, cutting delivery turnaround ~50% - directly answering the JD's repeated call to "actively use AI to streamline existing processes."
- Applied anomaly detection to identify cost-leakage patterns in operational datasets (~15% client savings); built a predictive ML model for chemical consumption at an industrial client, using process-level operational data to forecast production inputs - maintenance-adjacent predictive analytics on real industrial data.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 - Nov 2023 | Melbourne, Australia

- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders; presented system architecture and outcomes to cross-functional teams.
- Built a distributed anomaly-detection pipeline (Databricks, PySpark) to surface data-quality issues in government procurement datasets, improving detection accuracy ~20%.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 - Jul 2021

- Analysed blast-wave datasets using statistical models and Butterworth filtering, improving noise-level prediction accuracy ~20% and refining defence-equipment designs ~10%.
- Compiled findings into a 70-page technical report (MATLAB, Excel), communicating quantitative results to research stakeholders.

PROJECTS

Global Multi-Region Spend Analytics Platform (SQL Server, Python, Power BI, Azure)

- Designed and owns the full analytics pipeline for a global spend dataset (~6M rows, 13 sub-regions); built automated data ingestion, a ~11–12k-rule spend-categorisation engine, multi-language data cleaning, advanced spend analysis (rolling windows, PO splitting), and operational dashboards. Saved ~3–3.5 hrs/week versus prior manual process; demonstrates the exact work-order-data + multi-provider standardisation pattern required in Works Delivery analytics.

Cloud Data Platform - Snowflake + Power BI (AWS, Terraform, Snowflake, Power BI)

- Provisioned full cloud infrastructure as code (S3, IAM, EC2); built automated API-to-Snowflake ingestion pipeline and connected Power BI reporting layer; modular multi-environment design (dev/staging/prod) for reproducible deployments.

End-to-End Analytics & Decision Support System (Python, AI automation, forecasting, BI)

- Integrated 8+ data sources into a centralised analytics platform; built AI-driven workflows for automated reporting and insight generation, short-term operational forecasting, and performance benchmarking against external indicators - in ongoing client use.

Multi-Agent Data QA Validation Workflow (Claude, Python, agentic AI)

- Designed a 5-agent + orchestrator pipeline to validate client datasets prior to delivery; caught data integrity issues systematically, raising consistency and quality of analytics outputs across enterprise clients.

EDUCATION

Master of Data Science, Monash University | Feb 2022 - Dec 2023

Statistics I & II, Machine Learning, Applied Forecasting, Communicating with Data, High Dimensional Analysis

Bachelor of Engineering, Thapar University | May 2017 - May 2021

Mathematics I/II/III, Statistics, Analog & Digital Electronics

Eligibility: Victorian Driver's Licence held. Available for travel to MW sites (ETP, WTP, Winneke, SERO, Docklands).